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Project Title:	Transcriptional Regulation and Evolution of Bilaterian Body Plans		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
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Contact Monitor:		**Contact Phone:**	
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Task Description:	The fundamental mechanism by which animal forms evolve is change in the genomic regulatory networks that control the developmental processes by which these forms are built in each life cycle: understanding evolution of animal forms comes down to understanding evolution of gene networks. But this is a yet unsolved problem because genomic regulatory networks are just beginning to be solved at the DNA level. We are in an unparalleled position to attack this problem because we have in hand what may be the first example of such a network, viz a network that underlies endomesoderm development in sea urchin embryos. We propose to perform a comparative analysis of network architecture in a starfish, the embryos of which appear to produce endomesoderm in a somewhat similar way. However, the last common ancestor of starfish and sea urchins lived >500 mya. A large amount of preliminary work has already been done: we have arrayed Asterina miniata (Am) BAC and embryo cDNA libraries; many endomesoderm regulatory		

	genes are already isolated and their expression characterized; and most importantly, we have developed an excellent method for gene transfer into Am eggs for study of gene regulation in development. Thus we are poised to carry out the first study of regulatory gene network evolution. Preliminary results suggest that many deeply informative surprises await us.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will help in understanding planetary bioscience.
Task Progress:	We are undertaking a comparative analysis of gene regulatory network (GRN) architectures governing endomesoderm development in sea urchins and starfish. We are in an unparalleled position to make such a comparison because the gene regulatory network governing this process in the sea urchin is extremely well known. The objective of this project is to determine orthologuous regions of this network from the starfish, Asterina miniata and thence to understand which processes are conserved (indicative an essential feature of their development), and how evolutionary differences are incorporated in gene networks. In our previous report we detailed the first set of immensely revealing comparisons. We showed that there was a most remarkable homology between sea urchin and starfish GRNs in that an identical 3-gene reinforcing loop was found exactly conserved in the two taxa. The conservation of this feature across the immense period of evolutionary time such as separates these echinoderms (~500 MY) demonstrates the indispensable nature of such a process in their development. Since the publication of this work we have extended our comparison of GRN architectures in the two taxa (Fig. 1). We examined processes that, in the sea urchin are governed by the delta signal, viz the micromere-derived delta signal which specifies adjacent cells to mesoderm, and then the later signal of delta in the mesoderm to specify adjacent endoderm. Starfish develop without micromeres and hence without the delta signal that is essential in sea urchins for the correct specification of mesodermal cell types. To undertake this new comparative aspect we isolated and characterized several regulatory genes from the starfish, i.e., delta, gcm and gatac. We also isolated and characterized the starfish ortholog of the sea urchin Hnf6 gene. In the sea urchin, gcm responds to the micromere derived Delta signal to consequently specify mesodermal pigment cells. To our surprise, gcm is expressed very similarly in the later embryos of starfish and sea urchins (Fig.2), although starfish do not develop pigment cells. We further show that gcm is not regulated by Delta in the starfish and does not have an early vegetal plate restricted expression (Fig.3). Furthermore, none of the regulatory genes that we know to have mesodermally restricted expression patterns in the starfish require Delta for their activation. Thus, the micromere-specific function of Delta is absent in starfish. Delta is required in both taxa however for correct expression of the endodermally expressed gatae gene demonstrating that the later function of Delta is conserved. These analyses are revealing of how evolutionary changes are incorporated into network architectures. This year we have also made great progress in deciphering the cis-regulatory architecture of some of the regulatory genes that occupy key nodes in the starfish network. Even over such long evolutionary time as exists between starfish and sea urchins, coding sequences are often well conserved while essentially no common identity is apparent in regulatory regions. Still, conservation of transcription factor regulatory interactions (such as those found in conserved regions of regulatory architecture) implies that conservation of some sort must exist in cis-regulatory DNA. We have isolated cis-regulatory modules that recapitulate correct expression of the brachyury, otx and delta genes. The ~500 bp module regulating brachyury expression is located proximal to the site of transcription initiation. In early development, this module precisely directs reporter gene expression to the edge of the blastopore. More distal modules, downstream of the coding sequence, are also required to enhance the level of reporter expression. This is entirely different to the arrangement of modules known to regulate the brachyury gene in the sea urchin. This is although these genes are regulated by the same factors and are both expressed in the edge of the blastopore in each embryo. We are now undertaking more detailed analyses of binding site arrangements in these modules. We also isolated an ~500 bp module that recapitulates the correct expression of the otx gene in starfish. As is the case with the regulation of brachyury, the location of this module relative to the otx coding sequence that it regulates is very dissimilar to the location of the corresponding module in the sea urchin. The module is proximal to the start of transcription in the sea urchin and around 20 Kb downstream of the coding sequence in A. miniata. For otx however, we know that the distribution of binding sites between S. purpuratus and A. miniata is remarkably similar (Fig.4). As we analyze these two examples further we will learn how macro-evolutionary processes constrain binding site arrangement while allowing dispersal of entire modules. To further aid with isolating regulatory information from among genomic sequence, we constructed a large insert (average size of 140 Kb) BAC genomic library from a related starfish Dermasterias imbricata. Genomic sequence from these two species of starfish can be compared to reveal stretches of conserved non-coding of sequence, likely to represent cis-regulatory modules. Such comparative sequence analyses are now commonly used with high success for locating cis-regulatory modules. We also constructed BAC GFP recombinants for the starfish delta and brachyury genes. This is a technique whereby the first exon of the endogenous gene that is contained within a BAC clone is replaced by the sequence of the reporter gene GFP. Importantly this allows perturbation assays to be performed in the context of the complete cis-regulatory architecture of either the delta or brachyury gene. For instance, we are now able to delete the distal enhancer modules from the brachyury BAC GFP recombinant and determine the effect on expression of the reporter. This will allow us to better understand the function of these distal enhancer modules.
Bibliography Type:	Description: (Last Updated: 07/18/2007)
Articles in Peer-reviewed Journals	Otim, O, Hinman, V. F. and Davidson, E. H. "Expression of AmHnf6, a starfish ortholog of a transcription factor with multiple distinct roles in sea urchin development." Gene Exp. Patterns. In press , Jan-2004
Articles in Peer-reviewed Journals	Hinman, V. F., Nguyen, A. T., Cameron, R. A., and Davidson, E. H. "Developmental gene regulatory network architecture across 500 MY of echinoderm evolution. " Proc. Natl. Acad. Sci. USA 100, 13356-13361. , Aug-2003
Books/Book Chapters	Hinman, V. F., and Davidson, E. H. "System-level Properties Revealed by a Gene Regulatory Network Analysis of Pregastrular Specification in Sea Urchins. " in "Gastrulation’" Ed. C. Stern. Cold Spring Harbor Laboratory Press. Cold Spring Habor, NY., Jul-2004